

Online learning: Adaptive Hedge

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1 Introduction

1.1 Project description

We implement and empirically compare two algorithms for the prediction-with-expert-advice problem: *Hedge* with a range of fixed learning rates, and *AdaHedge* (van Erven et al., 2011). Both algorithms are implemented in Python.

The algorithms are evaluated on three synthetic loss-sequence regimes chosen to highlight complementary behaviours:

- a *stochastic* regime, with a single consistently better expert;
- an *adversarial* regime, in which the best expert shifts between contiguous segments;
- a *low-gap* regime, in which no expert is meaningfully better than the others.

The adversarial regime is run at two distinct hardness levels to investigate whether and how the relative performance of algorithms changes with shift frequency. Each regime is run at $K = 10$ experts. For every regime we display the cumulative loss, the average loss per round, and the regret of each algorithm. In the final part, we trace AdaHedge’s effective learning rate η_t to show the adaptation behaviour and its dependence on the current data.

1.2 Project structure

```
src/  
  algorithms.py      Hedge and AdaHedge implementations  
  losses.py          synthetic loss-sequence generators  
  experiments.py     run loops, regret computation, plotting helpers  
notebooks/  
  experiments.ipynb  reproduces every experiment, saves the figures  
report.pdf
```

The notebook coordinates the experiments. All algorithmic logic is contained in `src/algorithms.py`, all loss-sequence generators in `src/losses.py`, and all aggregation and plotting in `src/experiments.py`.

1.3 Usage

```
python3 -m pip install -r requirements.txt  
jupyter notebook notebooks/experiments.ipynb
```

Complete execution should take less than a minute.

2 Experimental setup

2.1 Loss-sequence regimes

The three regimes follow van Erven et al. (2011).

Stochastic. A single fixed “good” expert (expert 0) with loss $\ell_{t,0} \sim \text{Bernoulli}(0.3)$, and all others with loss $\ell_{t,k} \sim \text{Bernoulli}(0.5)$, independently across rounds.

Adversarial (shifting best). The horizon is partitioned into n contiguous segments of length T/n . In segment i , expert $i \bmod K$ suffers loss 0.1 and all others suffer loss 0.9. The sequence is deterministic, but no fixed expert is uniformly good. We evaluate this regime at two hardness levels:

- $n = 2$: a single midpoint shift, with two segments of $T/2 = 1000$ rounds.
- $n = 5$: four shifts over the horizon, with five segments of $T/5 = 400$ rounds.

Low-gap. Expert 0 suffers constant loss 0.49; all others suffer constant loss 0.5. The gap is 0.01 per round and the sequence is deterministic.

2.2 Hyperparameters

We fix $K = 10$ experts at horizon $T = 2000$, and compare AdaHedge against four Hedge variants:

$$\eta \in \{0.001, \eta^*/4, \eta^*, 4\eta^*\}, \quad \eta^* = \sqrt{8 \ln K/T} \approx 0.096.$$

Choice of η . The four values span approximately two orders of magnitude and surround η^* on both sides. $\eta = 0.001$ is deliberately small (nearly two orders of magnitude below η^*) so that the cost of severe under-tuning is observable. $\eta = \eta^*/4$ is moderately undersized. η^* is the worst-case-optimal value. $\eta = 4\eta^*$ is moderately oversized. Together they expose the sensitivity of fixed- η Hedge on either side of η^* and identify the side on which benign regimes favour deviation from the worst-case rate.

Choice of K and T . We fix $K = 10$ rather than sweeping over a range. The expert-pool size enters the regret bound only through the mild $\sqrt{\ln K}$ factor, and some exploratory runs confirmed that the qualitative ranking of the algorithms in all four regimes is unchanged in the range $K \in \{5, 20\}$, so, varying K adds complexity without contributing to the comparison. $K = 10$ is representative — the problem is non-trivial (uniform play is far from optimal) and large enough that the $\sqrt{\ln K}$ factor is meaningful. The horizon must be long enough for the cumulative-regret curves to leave the small- t transient regime.

Reproducibility. All experiments use `numpy's default_rng` with seeds in $[0, N_{\text{seeds}})$. The stochastic regime is repeated over 20 independent seeds; the adversarial ($n = 2, n = 5$) and low-gap settings are deterministic and run once each. Within a given seed all algorithms share the same loss sequence, so the standard-deviation bands in the stochastic figure reflect only the variability induced by the seeds.

2.3 Metrics and visualisation

For every regime we display three panels.

- **Cumulative loss L_t :** tracing the total loss incurred by each algorithm, and the best expert in hindsight.
- **Average loss per round L_t/t :** bounded in $[0, 1]$, exposing each algorithm's asymptotic per-round loss directly. We decided to include this view since often the cumulative loss curves of the algorithms are visually indistinguishable due to the large horizon and the small per-round differences.
- **Regret R_t on a *symlog* y-axis:** the gap between each algorithm's cumulative loss and that of the best expert in hindsight.

Each regime figure is a 1×3 grid (cumulative loss, average loss, regret). Shaded bands are ± 1 standard deviation across seeds. Hedge variants use a blue gradient ordered by η (light = small, dark = large), while AdaHedge is drawn as a thicker red line.

3 Implementation details

The source lives under `src/` and is split into three modules. We summarise each below, then expand on two non-obvious design choices in `algorithms.py`.

src/losses.py. Synthetic loss-sequence generators, one per regime: `stochastic_losses`, `adversarial_shifting`, and `low_gap_losses`. Each returns a (T, K) array of losses in $[0, 1]$.

src/algorithms.py. Implements `Hedge` and `AdaHedge` as classes sharing a `predict()/step(loss)` interface. `Hedge` takes a constant learning rate η as a constructor argument; `AdaHedge` maintains the cumulative mixability gap Δ_t internally and recomputes $\eta_t = \ln K / \Delta_{t-1}$ at every round. Both algorithms operate entirely in the log domain: the per-expert cumulative loss vector L_t is stored, the unnormalised log-weights $-\eta L_{t-1}$ are formed lazily inside `predict()`, and `AdaHedge`’s mix loss is evaluated via `logsumexp`. Two aspects of this implementation warrant detailed discussion.

Numerical stability of the softmax and the mix loss. The unnormalised log-weights $-\eta L_{t-1,k}$ drift to large negative values as cumulative losses accumulate: at $\eta = 0.1$ and $T = 2000$ they reach ≈ -200 , where $e^{-200} \approx 10^{-87}$ underflows silently.

src/experiments.py. This coordinates the experiments. `run_with_seeds` accepts a loss generator and a dictionary of algorithm factories, generates the loss sequence once per seed, and feeds that sequence to every algorithm. `cumulative_regret` computes $R_t = H_t - \min_k L_{t,k}$ from the per-round learner losses and the loss matrix. Also, there are three plotting helpers (`plot_cumloss`, `plot_avg_loss`, `plot_regret`).

4 Results

Table 1 reports the final regret R_T for every algorithm in every regime, averaged over seeds where applicable. The Hedge worst-case bound is $\sqrt{T \ln K / 2} \approx 48$ at $K = 10$.

algorithm	stochastic	adversarial $n=2$	adversarial $n=5$	low-gap
Hedge $\eta = 0.001$	355.6	580.9	147.3	18.0
Hedge $\eta = \eta^*/4$	97.8	67.6	29.7	17.5
Hedge $\eta = \eta^*$	26.0	17.3	8.1	15.2
Hedge $\eta = 4\eta^*$	8.2	4.8	2.7	6.0
AdaHedge	5.9	1.1	0.8	0.01

Table 1: Final regret R_T at $T = 2000$, $K = 10$, for every algorithm $\times$ regime combination. Stochastic: mean over 20 seeds. All other settings: single deterministic run. Bold marks the best algorithm per column. The Hedge worst-case bound $\sqrt{T \ln K / 2} \approx 48$.

4.1 Stochastic regime

The stochastic regime provides the clearest illustration of the value of adaptivity (Figure 1). Hedge with $\eta = 0.001$ acts almost like a uniform policy and reaches $R_T \approx 356$, well above the worst-case bound, as expected of a severely mistuned learner. As η grows the final regret

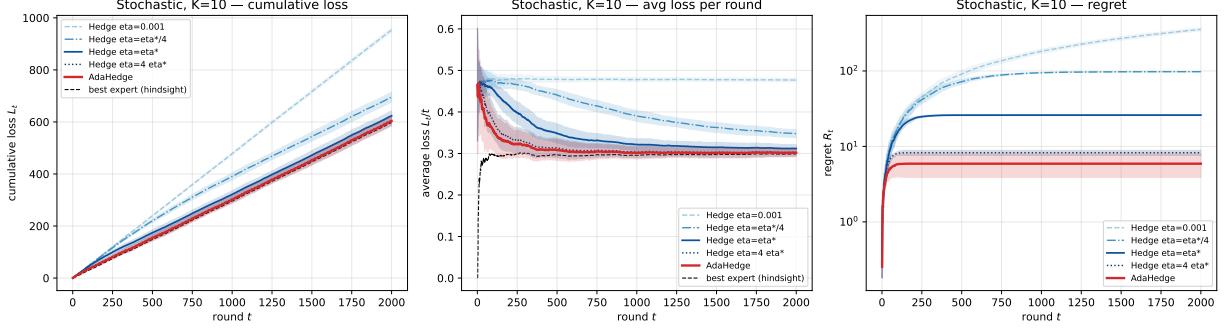


Figure 1: Stochastic regime ($N = 20$ seeds, $K = 10$). Left: cumulative loss; the dashed black curve is the best expert in hindsight. Middle: average loss per round. Right: regret R_t on a symlog axis. Shaded bands are ± 1 std.

decreases monotonically, and the over-tuned $4\eta^*$ already attains $R_T \approx 8$: on stochastic data the worst-case-optimal η^* is conservative.

AdaHedge attains the smallest final regret ($R_T \approx 6$) without tuning. The middle panel shows its average loss converging to the best-expert asymptote 0.3 faster than any fixed- η Hedge. Section 4.4 shows that AdaHedge does so by selecting an effective η_t much larger than η^* , which proves to be effective in this regime.

4.2 Adversarial regime: shifting best expert

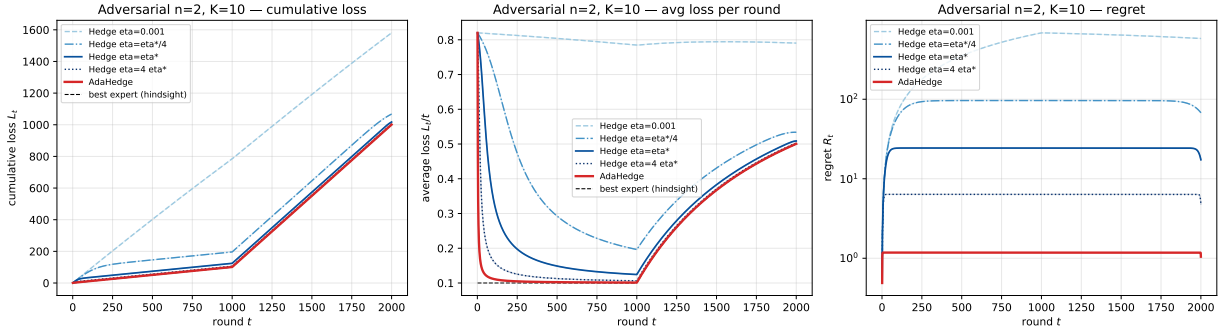


Figure 2: Adversarial regime, $n = 2$ ($K = 10$). A single shift at the midpoint, two segments of $T/2 = 1000$ rounds.

$n = 2$. Figure 2 corresponds to a single midpoint shift. AdaHedge attains $R_T \approx 1.1$, $4\eta^*$ Hedge attains ≈ 4.8 , the most mistuned variant ($\eta = 0.001$) reaches ≈ 581 . Hedge performance improves monotonically with η , and the worst-case-optimal η^* is again conservative ($R_T \approx 17$). Unlike in the $n = 5$ case, the staircase pattern in the regret panel is not visible here: the lines seem completely flat during all the central phase, apart from the variant with $\eta = 0.001$.

$n = 5$. Figure 3 exhibits the regret staircase that arises whenever the best expert in hindsight changes: at every segment boundary the learner continues to track the new leader, while the comparator only “catches up” during its own segment. The middle panel makes the shifting structure directly visible: the learner’s average loss dips towards 0.1 as it locks onto the active best expert, and recovers after each leader change. AdaHedge gets $R_T \approx 0.8$, roughly three times lower than the best fixed- η Hedge (≈ 2.7 for $4\eta^*$). Relative to $n = 2$, no fixed expert is best for long, so the comparator is weaker and absolute regrets shrink.

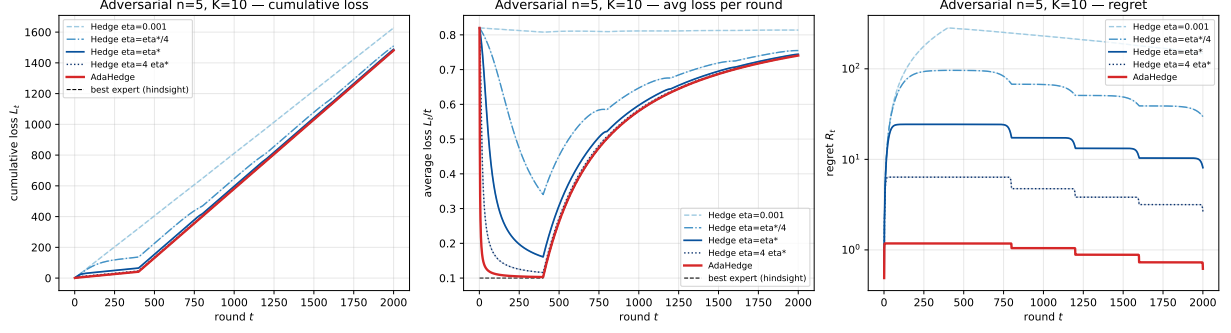


Figure 3: Adversarial regime, $n = 5$ ($K = 10$). The best expert shifts every $T/5 = 400$ rounds. The right panel exhibits the characteristic regret “staircase” as the regret of every algorithm dips at each of the four segment boundaries.

4.3 Low-gap regime

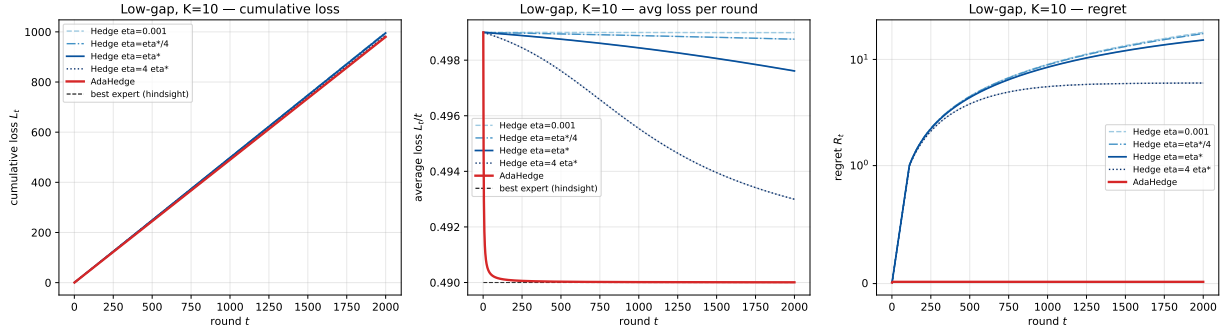


Figure 4: Low-gap regime ($K = 10$). Expert 0 suffers constant loss 0.49, all others 0.5; the sequence is deterministic.

In the low-gap regime (Figure 4), despite the small gap, the algorithms are clearly differentiated. The key is the absence of per-round noise: with constant losses, the mixability gap $\delta_t = h_t - m_t$ is positive from the very first round, so $\Delta_1 = 0.009$ and AdaHedge immediately sets $\eta_2 = \ln K / \Delta_1 \approx 256$. At that rate expert 0 receives overwhelming weight from round 2 onward, giving AdaHedge near-zero regret ($R_T \approx 0.01$). The overtuned $4\eta^*$ Hedge also concentrates quickly and attains $R_T \approx 6$. Conservative variants converge more slowly: $R_T \approx 15$ for η^* and ≈ 18 for $\eta^*/4$, reflecting how long each remains close to the uniform policy.

4.4 AdaHedge’s effective learning rate

Figure 5 reports the effective η_t selected by AdaHedge on each of the four loss sequences. Four observations:

- **Stochastic:** η_t remains above η^* for the entire horizon, decaying in the first part. AdaHedge identifies the easy regime and operates close to FTL.
- **Adversarial $n = 2$:** The segment remains stable for almost the entire horizon, above η^* .
- **Adversarial $n = 5$:** a similar overall behaviour, but with some visible drops in the right part of the graph (actually from before the half of the horizon) as the algorithm reacts to the shifts.
- **Low-gap:** after a single FTL round, η_t settles to ≈ 256 and stays there for all the computation. AdaHedge operates close to FTL for the entire horizon, much above η^* .

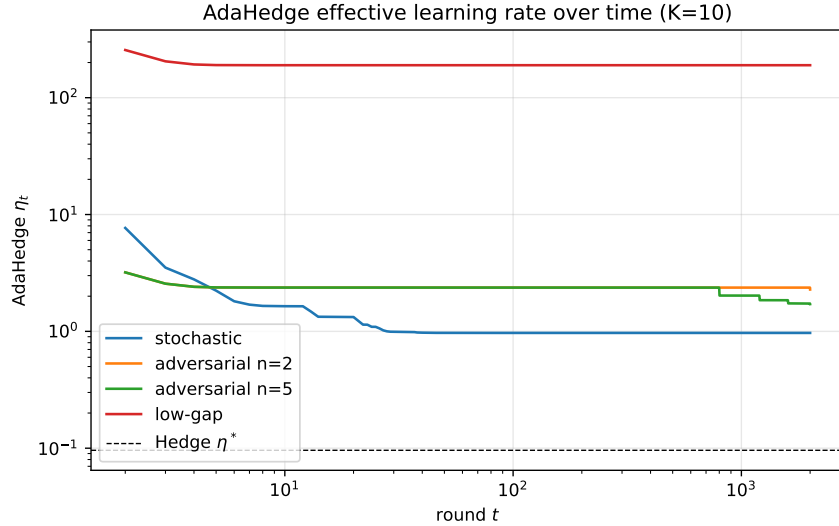


Figure 5: AdaHedge’s effective learning rate $\eta_t = \ln K / \Delta_{t-1}$ plotted on a log-log axis, on the four loss sequences studied in Section 4. The two adversarial settings are shown as independent traces. Dashed: the horizon-aware Hedge optimum η^* .

The same algorithm, on four loss sequences, settles on four distinct learning-rate trajectories — direct evidence that the adaptation is non-trivial.

5 Discussion

AdaHedge outperforms every fixed- η Hedge in all four settings, with margins that vary with the structure of the regime. In the stochastic regime the advantage is moderate ($R_T = 5.9$ vs 8.2 for the best Hedge). In the adversarial settings it is substantial (1.1 vs 4.8 at $n = 2$; 0.8 vs 2.7 at $n = 5$). In the deterministic low-gap regime it is even more pronounced (0.01 vs 6.0): without per-round noise, the mixability gap grows from the very first round, giving AdaHedge an effectively large η from round 2 onward and near-FTL performance throughout. In every case the improvement comes without any horizon or regime information.

One practical lesson is the cost of mistuning Hedge. A factor-of-four undersizing of η already multiplies regret by $\approx 4\times$ in the stochastic regime, and $\eta = 0.001$ multiplies it by more than an order of magnitude. AdaHedge removes the tuning decision entirely.

References

Tim van Erven, Peter D. Grünwald, Wouter M. Koolen, and Steven de Rooij. Adaptive hedge. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 24, 2011.

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